



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 4-Solutions

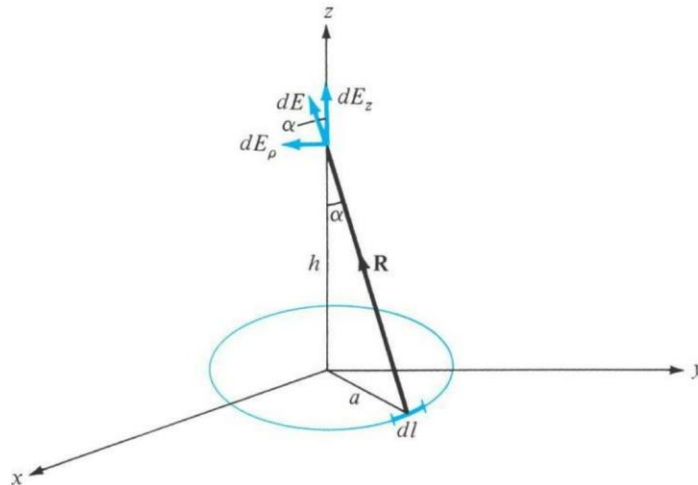
1. A circular ring of radius a carries a uniform charge ρ_L C/m and is placed on the xy -plane with axis the same as the z -axis. Show that:

$$\mathbf{E}(0, 0, h) = \frac{\rho_L a h}{2\epsilon_0 [h^2 + a^2]^{3/2}} \mathbf{a}_z$$

- a. What values of h give the maximum value of $\mathbf{E}$?
- b. If the total charge on the ring is Q , find $\mathbf{E}$ as $a \rightarrow 0$.

Solution

Consider the system as shown in the following figure.



Using the equation:

$$\mathbf{E} = \int \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$

and deriving each term in the equation. In this case:

$$d\mathbf{l} = a d\phi \mathbf{a}_\phi$$

$$\mathbf{R} = a(-\mathbf{a}_\rho) + h \mathbf{a}_z$$

$$R = |\mathbf{R}| = [a^2 + h^2]^{1/2}$$

$$\mathbf{a}_R = \frac{\mathbf{R}}{R}$$

$$\frac{\mathbf{a}_R}{R^2} = \frac{\mathbf{R}}{|\mathbf{R}|^3} = \frac{-a \mathbf{a}_\rho + h \mathbf{a}_z}{[a^2 + h^2]^{3/2}}$$

Hence:

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_o} \int_{\phi=0}^{2\pi} \frac{(-a\mathbf{a}_\rho + h\mathbf{a}_z)}{[a^2 + h^2]^{3/2}} a d\phi$$

By symmetry, the contributions along $\mathbf{a}_\rho$ add up to zero. This is evident from the fact that for every element $d\mathbf{l}$ there is a corresponding element diametrically opposite to it that gives an equal but opposite dE_ρ so that the two contributions cancel each other. Thus, we are left with the z-component. That is:

$$\mathbf{E} = \frac{\rho_L a h \mathbf{a}_z}{4\pi\epsilon_o [a^2 + h^2]^{3/2}} \int_{\phi=0}^{2\pi} d\phi = \frac{\rho_L a h \mathbf{a}_z}{2\epsilon_o [a^2 + h^2]^{3/2}}$$

As required.

1.

$$\frac{d|E|}{dh} = \frac{\rho_L a}{2\epsilon_o} \left\{ \frac{[a^2 + h^2]^{\frac{3}{2}}(1) - \frac{3}{2}(h)2h[a^2 + h^2]^{1/2}}{[a^2 + h^2]^{3/2}} \right\}$$

For maximum $\mathbf{E}$,

$$\frac{d|E|}{dh} = 0$$

which implies that:

$$[a^2 + h^2]^{1/2}[h^2 + a^2 - 3h^2] = 0$$

$$a^2 - 2h^2 = 0$$

$$h = \pm \frac{a}{\sqrt{2}}$$

2. Since the charge is uniformly distributed, the line charge density:

$$\rho_L = \frac{Q}{2\pi a}$$

so that:

$$\mathbf{E} = \frac{Qh}{4\pi\epsilon_0[a^2 + h^2]^{3/2}} \mathbf{a}_z$$

As $a \rightarrow 0$:

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 h^2} \mathbf{a}_z$$

or in general:

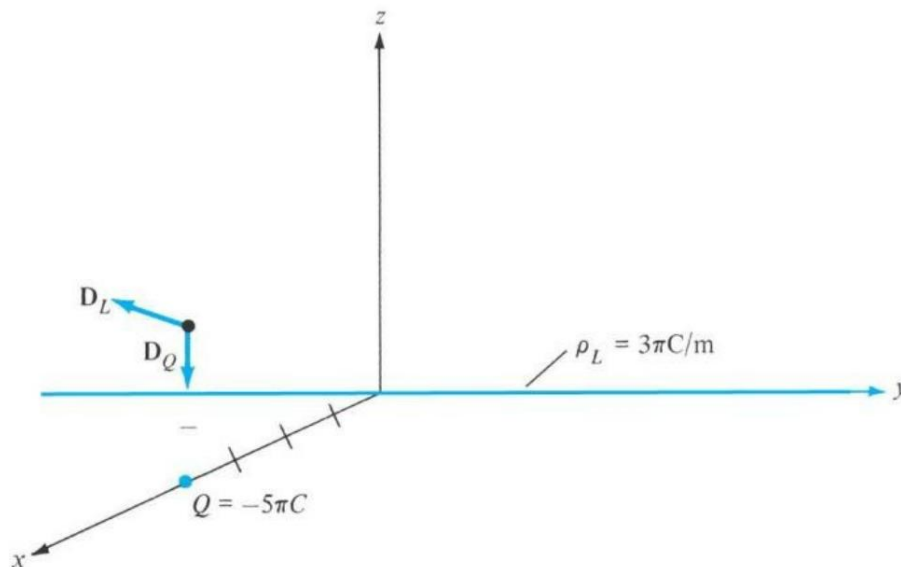
$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_R$$

which is the same as that of a point charge, as one would expect.

2. Determine $\mathbf{D}$ at $(4, 0, 3)$ if there is a point charge -5π mC at $(4, 0, 0)$ and a line charge 3π mC/m along the y -axis.

Solution

Let $\mathbf{D} = \mathbf{D}_Q + \mathbf{D}_L$, where $\mathbf{D}_Q$ and $\mathbf{D}_L$ are flux densities due to the point charge and line charge, respectively, as shown in the figure:



$$\mathbf{D}_Q = \varepsilon_o \mathbf{E} = \frac{Q}{4\pi R^2} \mathbf{a}_R = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi |\mathbf{r} - \mathbf{r}'|^3}$$

where $\mathbf{r} - \mathbf{r}' = (4, 0, 3) - (4, 0, 0) = (0, 0, 3)$.

Hence:

$$\mathbf{D}_Q = \frac{-5\pi \cdot 10^{-3}(0, 0, 3)}{4\pi |(0, 0, 3)|^3} = -0.139 \mathbf{a}_z \text{ mC/m}^2$$

also:

$$\mathbf{D}_L = \frac{\rho_L}{2\pi\rho} \mathbf{a}_\rho$$

In this case:

$$\mathbf{a}_\rho = \frac{(4, 0, 3) - (0, 0, 0)}{|(4, 0, 3) - (0, 0, 0)|} = \frac{(4, 0, 3)}{5}$$

$$\rho = |(4, 0, 3) - (0, 0, 0)| = 5$$

hence:

$$\mathbf{D}_L = \frac{3\pi}{2\pi(25)} (4 \mathbf{a}_x + 3 \mathbf{a}_z) = 0.24 \mathbf{a}_x + 0.18 \mathbf{a}_z \text{ mC/m}^2$$

Thus:

$$\mathbf{D} = \mathbf{D}_Q + \mathbf{D}_L = 240 \mathbf{a}_x + 41.1 \mathbf{a}_z \text{ } \mu\text{C/m}^2$$

3. For a spherical charge distribution:

$$\rho_v = \begin{cases} \rho_o(a^2 - r^2), & r < a \\ 0, & r > a \end{cases}$$

2. Find E and V for $r \geq a$.

3. Find E and V for $r \leq a$.

4. Find the total charge.

5. Show that E is maximum when $r = 0.745 a$.

Solution

For $r \geq a$:

$$Q_{enc} = \int \rho_v dv = \iiint \rho_o(a^2 - r^2)r^2 \sin \theta \, d\theta \, d\phi \, dr$$

$$Q_{enc} = \rho_o \int_0^\pi \sin \theta \, d\theta \int_0^{2\pi} d\phi \int_0^a (a^2 r^2 - r^4) \, dr$$

$$Q_{enc} = 4\pi\rho_0 \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right) \Big|_0^a = \frac{8\pi}{15} \rho_0$$

$$\Psi = \int \mathbf{D} \cdot d\mathbf{S} = \varepsilon_o E_r (4\pi r^2)$$

$$\Psi = Q_{enc}$$

$$\varepsilon_o E_r (4\pi r^2) = \frac{8\pi}{15} \rho_0$$

$$E_r = \frac{2\rho_0}{15\varepsilon_o r^2}$$

$$\mathbf{E} = \frac{2\rho_0}{15\varepsilon_o r^2} \mathbf{a}_r$$

$$V = - \int \mathbf{E} \cdot d\mathbf{l} = - \frac{2\rho_0}{15\varepsilon_o} \int r^{-2} \cdot dr = \frac{2\rho_0}{15\varepsilon_o r} + C_1$$

Since

$$V(r^- > 0) = 0$$

$$C_1 = 0$$

$$V = \frac{2\rho_0}{15\varepsilon_o r}$$

b. For $r \leq a$:

$$Q_{enc} = \rho_0 (4\pi) \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right) \Big|_0^r = 4\pi\rho_0 \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right)$$

$$E_r = \frac{Q_{enc}}{4\pi\varepsilon_o r^2} = \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r}{3} - \frac{r^3}{5} \right)$$

$$\mathbf{E} = \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r}{3} - \frac{r^3}{5} \right) \mathbf{a}_r$$

$$V = - \int \mathbf{E} \cdot d\mathbf{l} = - \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r^2}{6} - \frac{r^4}{20} \right) + C_2$$

$$V = \frac{\rho_0}{\varepsilon_o} \left(\frac{r^4}{20} - a^2 \frac{r^2}{6} \right) + C_2$$

Since $V(r = a^+) = V(r = a^-)$:

$$\frac{2\rho_0}{15\varepsilon_o a} = \frac{\rho_0}{\varepsilon_o} \left(\frac{a^4}{20} - \frac{a^4}{6} \right) + C_2$$

$$C_2 = \frac{2\rho_0}{15\varepsilon_o a} + \frac{7\rho_0 a^4}{60\varepsilon_o}$$

$$V = \frac{\rho_0}{\varepsilon_o} \left(\frac{r^4}{20} - a^2 \frac{r^2}{6} \right) + \frac{2\rho_0}{15\varepsilon_o a} + \frac{7\rho_0 a^4}{60\varepsilon_o}$$

c. The total charge is found in part (a) as:

$$Q = \frac{8\pi}{15} \rho_0$$

d. For $r \geq a$, E decays to zero with no maxima. For $r \leq a$:

$$E_r = \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r}{3} - \frac{r^3}{5} \right)$$

$$\frac{\partial E_r}{\partial r} = \frac{\rho_0}{\varepsilon_o} \left(\frac{a^2}{3} - \frac{3r^2}{5} \right) = 0$$

$$r = \frac{a\sqrt{5}}{3}$$

hence:

$$r = 0.7453 a$$